

Systematic evaluation of multifactorial causal associations for Alzheimer's disease and an interactive platform based on Mendelian randomization analysis — MRAD

Tianyu Zhao, Hui Li, Meishuang Zhang, Yang Xu, Ming Zhang, Li Chen 

Department of Pharmacology, College of Basic Medical Sciences, Jilin University, 126 Xinmin Street, Changchun City, Jilin Province, China, 130021 • Department of Neurology, Xuanwu Hospital, Capital Medical University, Beijing, China, 130021 • Neurology and Intracranial Hypertension & Cerebral Venous Disease Center, National Health Commission of China, Xuanwu Hospital, Capital Medical University, Beijing, China, 130021 • School of Nursing, Jilin University, 965 Xinjiang Street, Changchun City, Jilin Province, China, 130021

 https://en.wikipedia.org/wiki/Open_access

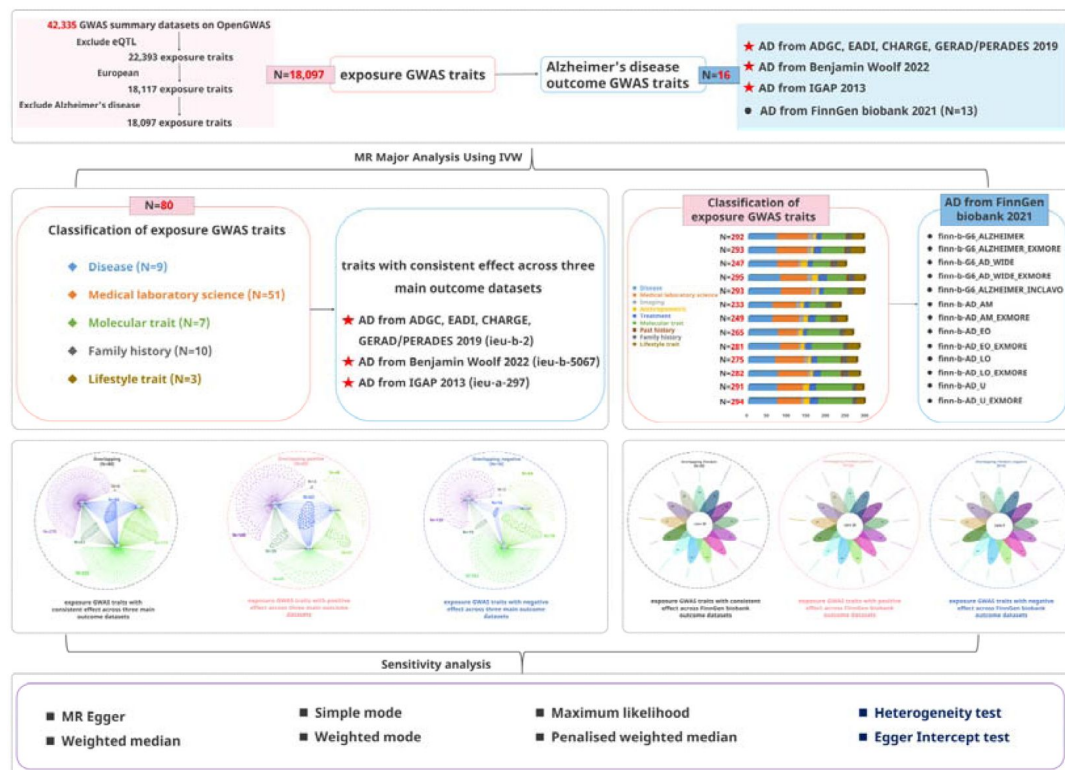
 Copyright information

Abstract

Alzheimer's disease (AD) is a complex degenerative disease of the central nervous system. Traditional epidemiological studies have reported several risk factors for AD. However, most epidemiological studies are insufficient to draw definitive conclusions on causal association due to the potential for reverse causality and confounding bias. Therefore, elucidating its pathogenesis remains challenging. Mendelian randomization (MR) was developed for assessing causality using genetic variants as a new approach in epidemiological research. In this study, we used MR analysis to investigate potential AD risk factors to support extensive AD research. We used the inverse-variance weighted (IVW) model as the major analysis method to perform hypothesis-free Mendelian randomization analysis on the data from MRC IEU OpenGWAS (18,097 exposure traits and 16 AD outcome traits), and conducted sensitivity analysis with six models, to assess the robustness of the IVW results, to identify various classes of risk or protective factors for AD, early-onset AD, and late-onset AD. We generated 400,274 data entries in total, among which the major analysis method of IVW model consists of 73,129 records with 4840 exposure traits, which fall into 10 categories: Disease (n=17,168), Medical laboratory science (n=15,416), Imaging (n=4,896), Anthropometric (n=4,478), Treatment (n=4,546), Molecular trait (n=17,757), Gut microbiota (n=48), Past history (n=668), Family history (n=1,114), and Lifestyle trait (n=7,038). For the convenience of display and operation, an online platform called MRAD has been developed using the Shiny package with MR analysis results. MRAD can be freely accessed online at <https://gwasmrad.com/mrad/>. Moreover, novel potential AD therapeutic targets (CD33, TBCA, VPS29, GNAI3, PSME1) are identified, among which CD33 was positively associated with the main outcome traits of AD, as well as with both EOAD and LOAD. TBCA and VPS29 were negatively associated with the main outcome traits of AD, as well as with both EOAD and LOAD. GNAI3 and PSME1 were negatively associated with the main outcome traits of AD, as well as with LOAD, but had no significant causal association with EOAD. This is one of the first and largest studies in this field. The findings of our research advance understanding of the etiology of AD.

Graphical Abstract

Identifying risk or protective factors for Alzheimer's disease based on Mendelian randomization



eLife assessment

This study introduces the MRAD database, which provides a **useful** tool for evaluating risk and protective factors for Alzheimer's disease through Mendelian randomization analysis. While the findings are supported by **solid** evidence, the study's value could be enhanced by addressing methodological concerns and ensuring rigorous validation of significant associations. The MRAD database has the potential to aid researchers and clinicians, but the current analysis appears **incomplete** without these refinements.

<https://doi.org/10.7554/eLife.96224.1.sa2>

Background

Alzheimer's disease (AD) is a progressive degenerative disease of the central nervous system, characterized by cognitive impairment, reduced functional capacity for daily living, and behavioral changes. It can be divided into two types: early-onset AD (EOAD, age of onset ≤ 65 years) and late-onset AD (LOAD, age of onset > 65 years); the proportion of LOAD in patients with AD is approximately 95%, with LOAD having a stronger genetic predisposition than EOAD^[1, 2, 3]. According to the latest data from the World Health Organization (WHO), the population with AD is currently over 50 million worldwide and is expected to rise to 115 million by 2050^[4, 5]. With

the increasing aging population, the incidence of AD continues to rise, making AD the fifth leading cause of death worldwide. Given that AD is a chronic complex disorder involving multiple pathophysiological changes, it is likely caused by the joint action of various factors in a multifaceted pathological process, and this intricate nature of AD contributes to the current challenges in its diagnosis and treatment, such as low consultation rates, high rates of misdiagnosis at initial consultations, and low rates of long-term standardized treatment^[6], thereby making AD one of the most perplexing diseases. Consequently, examining the pathogenic mechanisms of AD, identifying its risk factors, and conducting timely and effective early screening and diagnosis are of utmost importance.

Traditional epidemiological studies have reported common risk factors for AD. Some metabolic comorbidities are highly associated with AD, such as cardiovascular disease^[7,8], obesity^[9,10], and diabetes^[11,12]. Serological parameters such as C-reactive protein^[13], lipids^[14,15], and vitamin levels^[16–18] have been previously reported as potential biomarkers for AD. In addition, some factors related to lifestyle, family history, education, economic level, and environment correlate with AD^[19–22]. However, most epidemiological studies are insufficient to draw definitive conclusions on causal association due to the potential for reverse causality and confounding bias.

Mendelian randomization (MR) analysis is an emerging method to explore the causal association between AD and various factors^[23–25]. MR analysis reduces confounding and reverse causality due to the segregation and independent assortment of genes passed from parents to offspring^[26]. In the absence of pleiotropy (that is, genetic variation related to a disease via other pathways) and demographic stratification, MR can present a clear estimate of risk of disease^[27,28]. MR analysis is increasingly used to determine a causal relationship between potentially modifiable risk factors and outcomes^[29]. These advantages make MR a valuable tool to better elucidate the potential risk or protective factors for AD. But to date, AD has been reported as hypothesis driven MR study based on single factor, ignoring the potential role of a huge number of other risk factors. And also, due to the high degree of heterogeneity present in AD subtypes, which have different biological and genetic characteristics. Thus, the previous studies cannot offer a systematic and complete viewpoint.

To overcome these limitations, in this work we attempt to identify risk or protective factors causally associated with AD from a holistic and systematic perspective, thereby providing new ideas for understanding the AD pathogenesis, achieving early diagnosis, and developing clinical drugs. In the first place, this study uses a hypothesis free data mining approach to studying the possible etiology of Alzheimer's disease based on Mendelian randomization (MR), with specific attention to different AD subtypes (EOAD and LOAD). Based on this, we developed an online open integrated platform, MRAD (Mendelian randomization for Alzheimer's disease, <https://gwasmrاد.com/mrad/>). Moreover, the platform was further enriched by including related targets' information such as functions and pathways retrieved from the public database Uniprot. The platform is the first multi-dimensional, integrated, shared, and interactive comprehensive platform for AD MR research to date.

Methods

Database and software

The following databases and software packages were used in this study: MRC IEU OpenGWAS^[30] (<https://gwas.mrcieu.ac.uk/>), UniProt^[31] (<https://www.uniprot.org/>), EVenn^[32] (<http://www.ehbio.com/test/venn/#/>), R (version 4.1.2) software^[33].

MR design for AD (Figure S1)

Data sources

Exposure traits

Inclusion criteria: datasets of the European population. Exclusion criteria: (i) AD-related datasets.

In this study, 18,097 GWAS datasets were selected as exposure traits from the MRC IEU OpenGWAS database according to the above criteria. (See Table S1 for basic information.)

Outcome traits

Inclusion criteria: (i) datasets of patients with AD with complete information and clear data sources; (ii) datasets of the European population.

Exclusion criteria: (i) Number of SNPs <1 million; (ii) datasets with unspecified sex; (iii) datasets with a family history of AD; (iv) datasets with dementia.

Based on the above criteria, 16 GWAS datasets of outcome traits were selected from the MRC IEU OpenGWAS database, comprising datasets of AD from ADGC, EADI, CHARGE, GERAD/PERADES 2019 (ieu-b-2); AD from

Benjamin Woolf 2022 (ieu-b-5067); AD from IGAP 2013 (ieu-a-297) as the datasets of main outcome traits for AD, as well as 13 datasets from FinnGen biobank 2021 corresponding to various AD subtypes, referred to as AD-finn subtypes. (as shown in Figure S2)

Selection of instrumental variables

SNPs serve as instrumental variables for MR research. In this study, 18,097 exposures-variable SNPs were selected for MR research from the GWAS data (as mentioned in *Exposure traits*) respectively, with the selected SNPs fulfilling the following requirements: (i) a genome-wide significant association with risk factors ($p < 5 \times 10^{-8}$) in the European 1000 Genomes Project reference panel; (ii) independent of one another (that is, the r^2 of linkage disequilibrium (LD) is less than 0.001 within a 10,000-kb distance) to avoid potential biases caused by LD between SNPs in the analysis.

Statistical models for causal effect inference

A random-effects IVW model was used in this study as the major analysis method to uncover potential risk or protective factors for AD. An IVW model, based on the premise that all instrumental variables are valid, calculates the weighted average of the effect estimates of each instrumental variable by using the inverse of the variance of each instrumental variable as the weight, while not considering the intercepts in regression^[34]. To assess the robustness of the IVW results, sensitivity analysis was performed using six additional models: MR-Egger^[35,36], Weighted median^[37], Simple mode, Weighted mode, Maximum likelihood^[38], and Penalized weighted median, along with heterogeneity and horizontal pleiotropy assessment using the heterogeneity tests^[39] and Egger intercept tests^[40].

The above analyses were performed using the TwoSampleMR^[41] package in the R (version 4.1.2) software. Association of exposures with outcomes was assessed using odds ratio (OR) and 95% confidence interval (95% CI), with $OR > 1$ indicating a positive association (risk factor) and $0 < OR < 1$ indicating a negative association (protective factor). Differences with a two-sided $p < .05$ were considered statistically significant.

Building the MRAD platform

In this study, the online MRAD platform was developed using the Shiny package^[42] in R (version 4.1.2) and hosted on an Ubuntu 20.04 server. By leveraging Shiny, we combined the computational capabilities of R with modern web technologies, allowing to construct an interactive user interface with novel approaches.

Results

Results of hypothesis-free Mendelian randomization analysis for Alzheimer's disease

Based on hypothesis-free Mendelian randomization analysis for Alzheimer's disease, this study generated a total of 400,274 data points. The major analysis method of IVW model consists of 73,129 records with 4840 exposure traits, which fall into 10 categories: Disease (n=17,168), Medical laboratory science (n=15,416), Imaging (n=4,896), Anthropometric (n=4,478), Treatment (n=4,546), Molecular trait (n=17,757), Gut microbiota (n=48), Past history (n=668), Family history (n=1,114), and Lifestyle trait (n=7,038), as shown in Figure S3. To assess the robustness of the IVW results, sensitivity analysis was performed using six other models (MR-Egger with a total of 50,804 records, Weighted median with a total of 50,804 records, Simple mode with a total of 50,804 records, Weighted mode with a total of 50,804 records, Maximum likelihood with a total of 73,125 records, and Penalized weighted median with a total of 50,804 records).

MRAD platform integration

Based on the 400,274 data points stated above, we created herein is an online data analysis platform for identifying the risk or protective factors for AD called MRAD (Mendelian randomization for Alzheimer's disease, <https://gwasmrad.com/mrad/>). It contains six modules: (i) Home; (ii) Study Design; (iii) IVW interactive; (iv) IVW static; (v) Sensitivity analysis interactive; and (vi) Sensitivity analysis static; The platform provides a user-friendly search interface, allowing users to search, interactively visualize, analyze, and download the obtained results (see Supplementary Material for details). In our view, as the first interactive comprehensive platform for AD MR research to date, this online platform would benefit the field of scientific research in AD in numerous ways. On the one hand, it would allow researchers to quickly identify risk or protective factors from their own research and generate novel hypothesis regarding the molecular mechanism of AD. On the other hand, it would allow researchers with complementary expertise to provide multiple characterizations of the same data. As the platform is hosted on a server and accessed through a web interface, which could meet the multi-terminal compatibility, thereby MRAD's online presence could increase access to potential users.

MRAD utility data mining

To demonstrate the utility of MRAD platform, we focus on the IVW model-identified exposure traits that have significantly and consistently effect across three main outcome traits of AD to demonstrate the performance of the MRAD platform. Detailed investigation and reporting of other factors will be carried out in future research.

In this study, MR analysis was first performed on the three main outcome traits of AD to explore their potential risk or protective factors, leading to identification of a total of 80 exposure traits ($p < 0.05$), which fell into five Classification I categories: Medical laboratory science (n=51), Family history (n=10), Disease (n=9), Molecular trait (n=7), and Lifestyle trait (n=3). A total of 63 exposure traits (risk factors) were positively associated with all the three main outcome traits, while 16 exposure traits (protective factors) were negatively associated with the three main outcome traits,

with Ulcerative colitis (ebi-a-GCST000964) being negatively associated with the AD outcome traits of ieu-b-2 and ieu-a-297, and positively associated with the AD outcome traits of ieu-b-5067. MR analysis was performed on the outcome traits of 13 different AD-finn subtypes to further examine the causal association between the above-identified key common exposure traits and different subtypes of AD outcome traits. The results are provided below in detail.

Causal association between medical laboratory science and the main outcome traits of AD

In this study, the 51 medical laboratory science items that each had a causal effect on the main outcome traits of AD were grouped into three Classification II categories (blood lipids and lipoproteins (n=36), immunological tests (n=12), and plasma protein tests (n=3)).

1 Blood lipids and lipoproteins

A total of 36 blood lipids and lipoproteins items as exposure traits had effects on the main outcome traits of AD: (1) 32 of which were positively associated with the main outcome traits, 7 of which, e.g., apolipoprotein B (ieu-b-108), were positively associated with EOAD (finn-b-AD_EO) and LOAD (finn-b-AD_LO); free cholesterol in IDL (met-c-868) was positively associated with EOAD (finn-b-AD_EO); 4 of which, e.g., phospholipids in small LDL (met-d-S_LDL_PL), were positively associated with LOAD (finn-b-AD_LO), as shown in [Figure 1](#). (2) four of which were negatively associated with the main outcome traits, apolipoprotein A-I (ieu-b-107) was negatively associated with both EOAD (finn-b-AD_EO) and LOAD (finn-b-AD_LO), and the negative causal association was slightly stronger for EOAD than for LOAD; phospholipids to total lipids ratio in chylomicrons and extremely large VLDL (met-d-XXL_VLDL_PL_pct) was negatively associated with LOAD (finn-b-AD_LO). These findings are illustrated in [Figure 2](#).

2 Immunological tests

A total of 12 immunological test items as exposure traits had positive effects on the main outcome traits of AD. Six of which, e.g., CD33 on Monocytic Myeloid-Derived Suppressor Cells (ebi-a-GCST90001952), were positively associated with LOAD (finn-b-AD_LO), as shown in [Figure 3](#).

3 Plasma protein tests

A total of 3 plasma protein tests items as exposure traits had negative effects on the main outcome traits of AD. The three exposure traits were C-reactive protein (ukb-d-30710_raw, ukb-d-30710_irnt, and ieu-b-4764). All the three exposure traits were negatively associated with EOAD (finn-b-AD_EO) and LOAD (finn-b-AD_LO), as shown in [Figure 4](#).

Causal association between family history and the main outcome traits of AD

A total of 10 family history items as exposure traits had causal effects on the main outcome traits of AD. In particular, a parental or family history of AD increased the overall risk of developing AD, and was positively associated with both EOAD (finn-b-AD_EO) and LOAD (finn-b-AD_LO), as shown in [Figure 5](#).

Causal association between diseases and the main outcome traits of AD

In this study, the 9 diseases items that each had a causal effect on the main outcome traits of AD were grouped into four Classification II categories (dementia (n=5), neurodegenerative diseases (n=2), mental disorders associated with neurological diseases (n=1), and digestive system diseases (n=1)). Their causal effects with the main outcome traits of AD and the outcome traits of EOAD (finn-b-AD_EO) and LOAD (finn-b-AD_LO) are shown in [Figure 6](#).

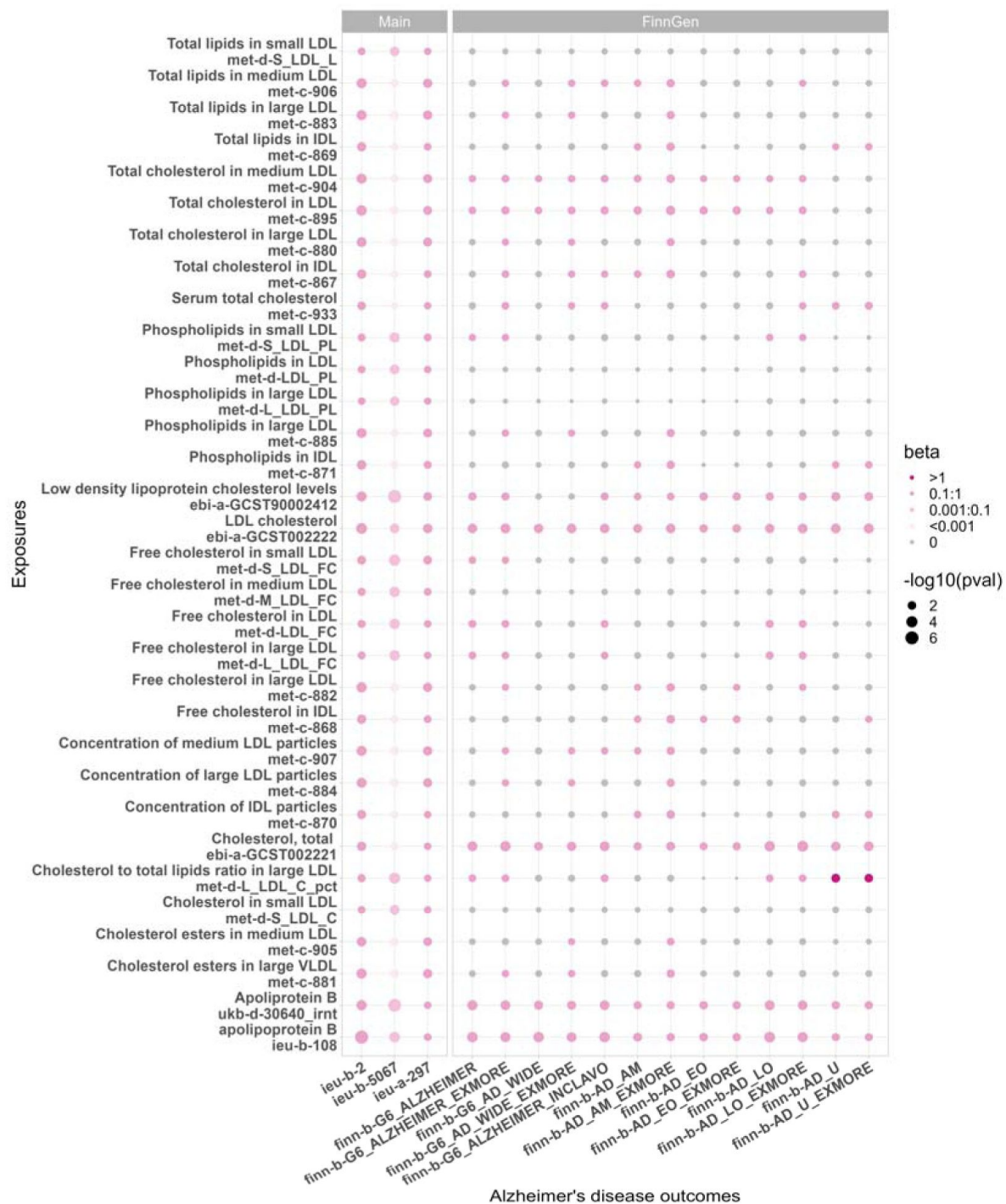


Figure 1.

Thirty-two blood lipids and lipoproteins items that were positively associated with the main outcome traits of AD.

Note: the pink dots in the figure represent positive association, with the color depth of the dots being positively proportional to the OR value (the darker the color, the larger the OR value), and the size of the dots being inversely proportional to the p-value (the smaller the p-value, the larger the dots). The gray dots represent no significant causal association ($p > 0.05$).

Figure 2.

Four blood lipids and lipoproteins items that were negatively associated with the main outcome traits of AD.

Note: the blue dots in the figure represent negative association, with the color depth of the dots being positively proportional to the OR value (the darker the color, the larger the OR value), and the size of the dots being inversely proportional to the p-value (the smaller the p-value, the larger the dots). The gray dots represent no significant causal association ($p > 0.05$).

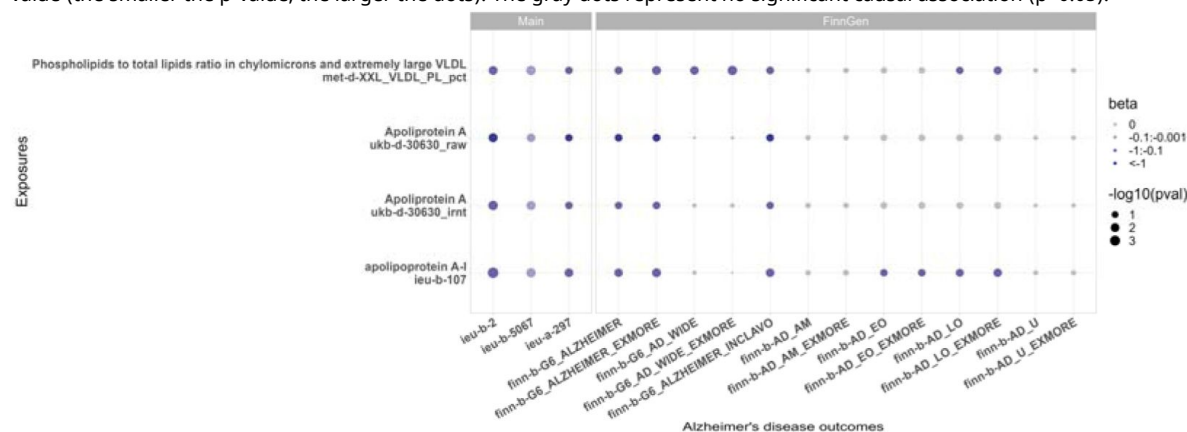


Figure 3.

Twelve immunological test items that were positively associated with the main outcome traits of AD.

Note: the pink dots in the figure represent positive association, with the color depth of the dots being positively proportional to the OR value (the darker the color, the larger the OR value), and the size of the dots being inversely proportional to the p-value (the smaller the p-value, the larger the dots). The gray dots represent no significant causal association ($p > 0.05$).

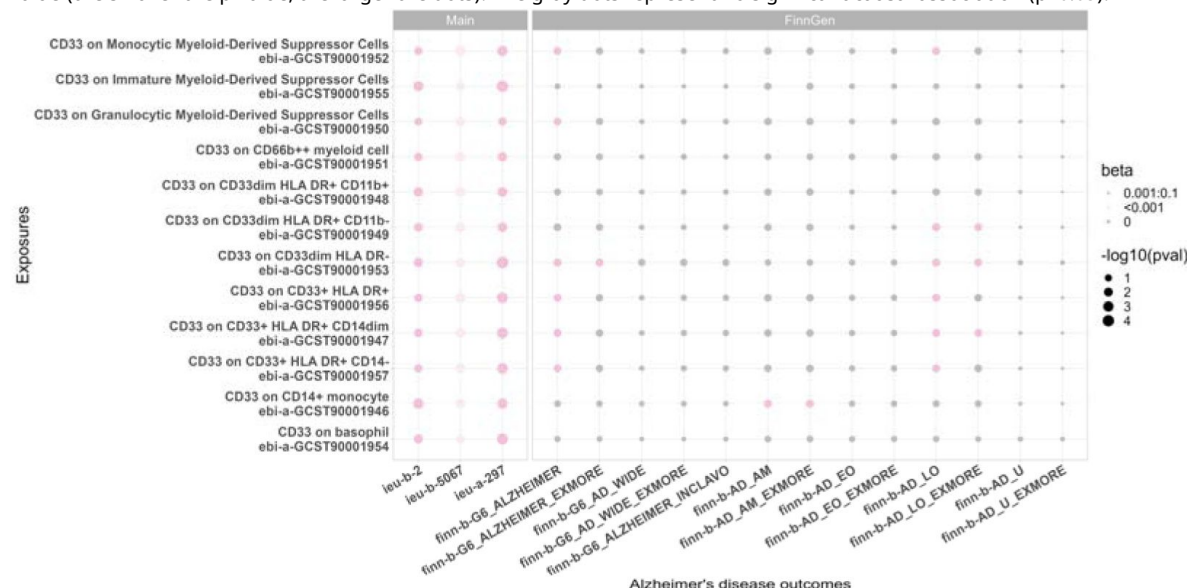


Figure 4.

Three plasma protein tests items that were negatively associated with the main outcome traits of AD.

Note: the blue dots in the figure represent negative association, with the color depth of the dots being positively proportional to the OR value (the darker the color, the larger the OR value), and the size of the dots being inversely proportional to the p-value (the smaller the p-value, the larger the dots). The gray dots represent no significant causal association ($p>0.05$).

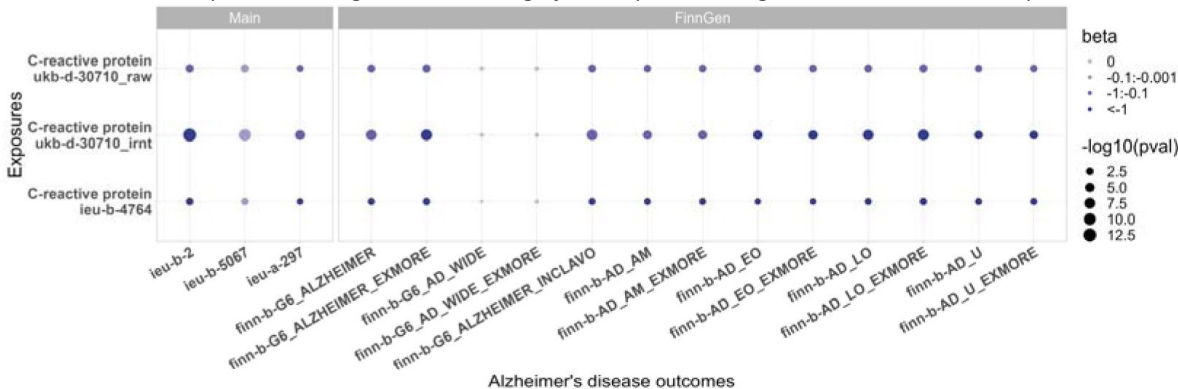
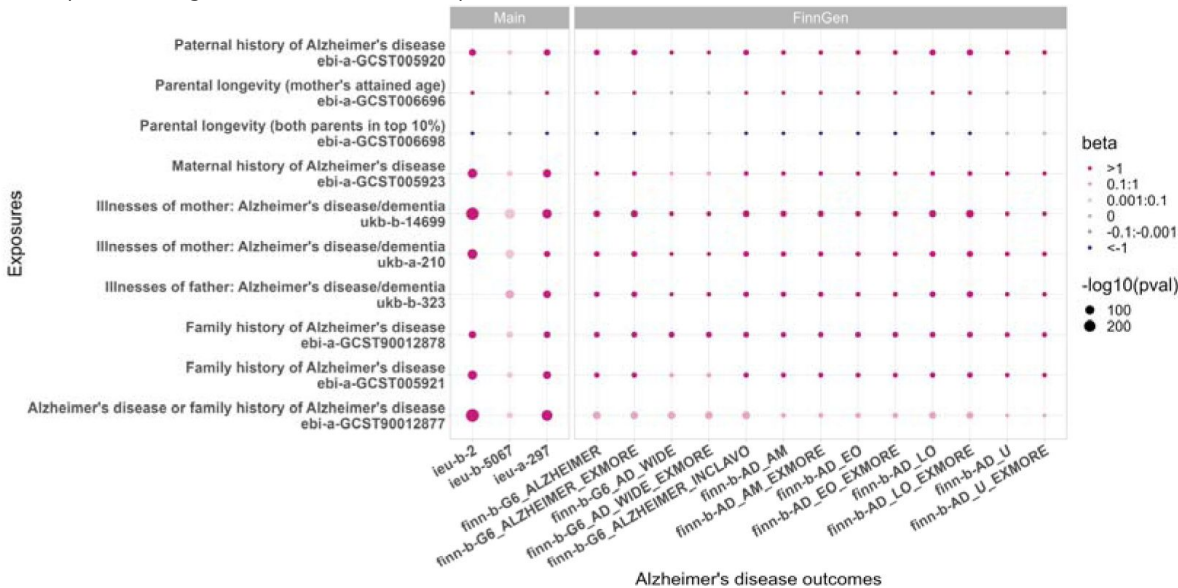


Figure 5.

Ten family history items with causal effects on the main outcome traits of AD.

Note: the pink dots in the figure represent positive association, the blue dots in the figure represent negative association, with the color depth of the dots being positively proportional to the OR value (the darker the color, the larger the OR value), and the size of the dots being inversely proportional to the p-value (the smaller the p-value, the larger the dots). The gray dots represent no significant causal association ($p>0.05$).



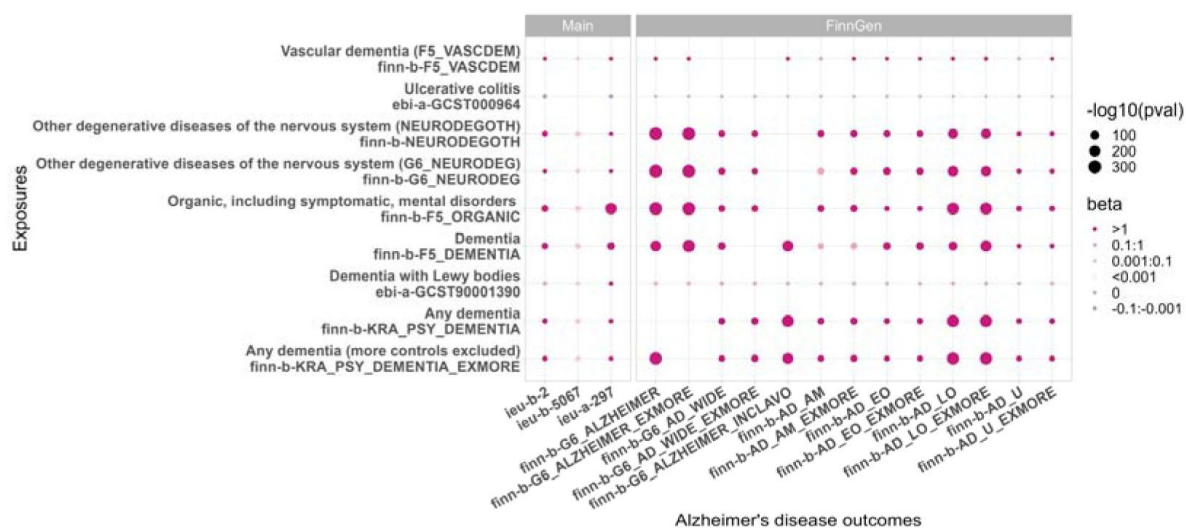


Figure 6.

Nine diseases items with causal effects on the main outcome traits of AD.

Note: the pink dots in the figure represent positive association, the blue dots in the figure represent negative association, with the color depth of the dots being positively proportional to the OR value (the darker the color, the larger the OR value), and the size of the dots being inversely proportional to the p-value (the smaller the p-value, the larger the dots). The gray dots represent no significant causal association ($p > 0.05$).

Causal association of molecular traits with the main outcome traits of AD

A total of 7 molecular trait items as exposure traits had causal effects on the main outcome traits of AD, among which Myeloid cell surface antigen CD33 (prot-a-439) was positively associated with the main outcome traits of AD, as well as with both EOAD (finn-b-AD_EO) and LOAD (finn-b-AD_LO). The remaining six were all negatively associated with the main outcome traits of AD, and their causal effects on the outcome traits of 13 AD-finn subtypes were as follows: (i) tubulin-specific chaperone A (TBCA; prot-a-2930) and vacuolar protein sorting-associated protein 29 (VPS29; prot-a-3203) were negatively associated with both EOAD (finn-b-AD_EO) and LOAD (finn-b-AD_LO); (ii) guanine nucleotide-binding protein G(k) subunit alpha (GNAI3; prot-a-1226) and proteasome activator complex subunit 1 (PSME1; prot-a-2420) were negatively associated with LOAD (finn-b-AD_LO), but had no significant causal association with EOAD (finn-b-AD_EO) ($p>0.05$); and (iii) neither glutamine (met-c-860) nor glutamine (met-d-Gln) had significant causal association with EOAD (finn-b-AD_EO) or LOAD (finn-b-AD_LO) ($p>0.05$), as shown in **Figure 7** [↗](#).

Causal association of lifestyle traits with the main outcome traits of AD

A total of 3 lifestyle trait items as exposure traits had causal effects on the main outcome traits of AD. Their causal effects with the main outcome traits of AD and the outcome traits of EOAD (finn-b-AD_EO) and LOAD (finn-b-AD_LO) are shown in **Figure 8** [↗](#).

Discussion

Despite decades of research on AD, controversy still remains regarding which factors play an important role in its pathogenesis. This study generated 400,274 data points based on hypothesis-free Mendelian randomization analysis for Alzheimer's disease, which provided a thorough and comprehensive evaluation with regard to risk or protective factors for AD. Based on this, for the convenience of display and operation, a user-friendly prediction platform was built online for MRAD. To our knowledge, this MR study covers most exposure traits that are causally associated with AD outcome traits, including diseases, medical laboratory science items, imaging items, anthropometric items, treatments, molecular traits, gut microbiota, past histories, family histories, and lifestyle traits, and reveals the causal associations between these exposure traits and different AD subtypes. The MRAD provides a one-stop online analysis service for researchers worldwide, including data retrieval → visualization → personalized analysis → data download. Users can obtain analysis results of different MR models (the main IVW model and six sensitivity analysis models) on 18,097 exposure traits and 16 AD outcome traits, totaling 400,274 records, and are allowed to set personalized parameters to meet different analysis needs. Additionally, the MRAD provides interactive visualization interfaces and download functions for the above results. This platform provides a unique resource for systematically identifying risk or protective factors of AD, which facilitates early identification, diagnosis, prevention, and treatment, with significant clinical and social value.

To briefly demonstrate the performance of MRAD, we explored the IVW model-identified exposure traits that had significantly consistently effect across all the three main outcome traits of AD.

The association of lipids and lipoproteins, C-reactive protein, family histories, neurological disorders, glutamine, and education level with AD has been widely reported^[23,43,66] and is consistent with the results of this study. Moreover, given that the prevalence of LOAD is about 95% in patients with AD and that LOAD has a stronger genetic predisposition than EOAD^[1,3], identifying new risk genes for LOAD is crucial for understanding its potential etiology. Therefore, this study further explored the relationships between these traits and different AD subtypes, leading to the following findings: (i) apolipoprotein B, cholesterol, total, LDL cholesterol, Low

Figure 7.

Seven molecular trait items with causal effects on the main outcome traits of AD.

Note: the pink dots in the figure represent positive association, the blue dots in the figure represent negative association, with the color depth of the dots being positively proportional to the OR value (the darker the color, the larger the OR value), and the size of the dots being inversely proportional to the p-value (the smaller the p-value, the larger the dots). The gray dots represent no significant causal association ($p>0.05$).

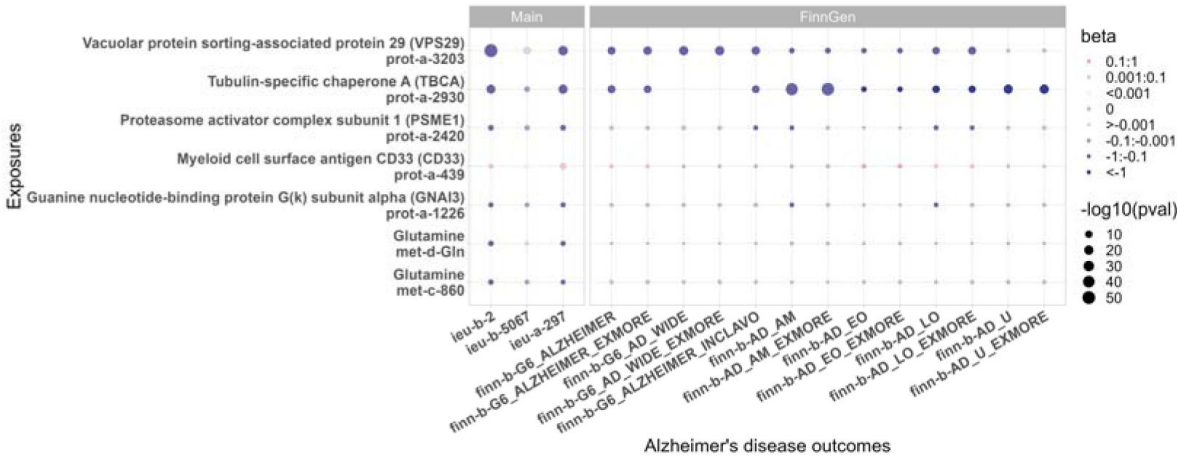
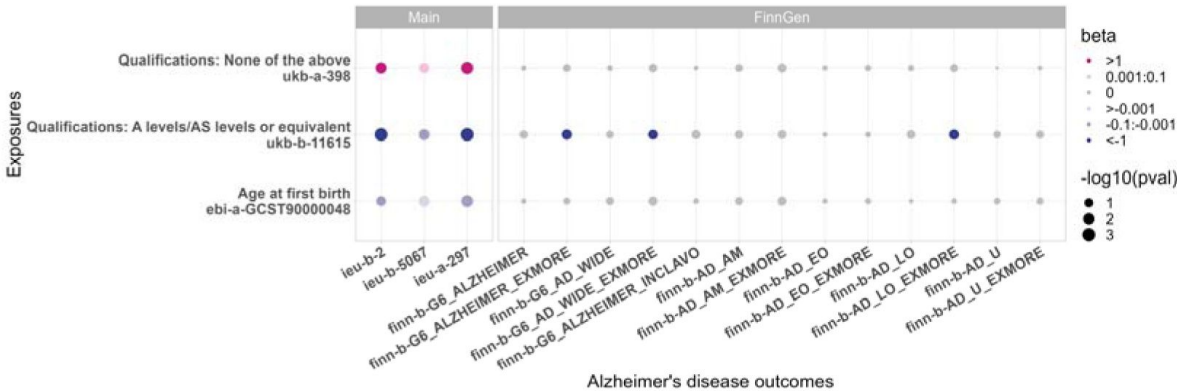


Figure 8.

Three lifestyle trait items with causal effects on the main outcome traits of AD.

Note: the pink dots in the figure represent positive association, the blue dots in the figure represent negative association, with the color depth of the dots being positively proportional to the OR value (the darker the color, the larger the OR value), and the size of the dots being inversely proportional to the p-value (the smaller the p-value, the larger the dots). The gray dots represent no significant causal association ($p>0.05$).



density lipoprotein cholesterol levels, total cholesterol in LDL, total cholesterol in medium LDL, cholesterol to total lipids ratio in large LDL, free cholesterol in large LDL, free cholesterol in LDL, phospholipids in small LDL, parental or family history of AD, parental longevity (mother's attained age), dementia, vascular dementia, dementia with Lewy bodies, other degenerative diseases of the nervous system, and organic, including symptomatic, mental disorders were all positively associated with LOAD; (ii) apolipoprotein A-I, phospholipids to total lipids ratio in chylomicrons and extremely large VLDL, C-reactive protein, parental longevity (both parents in top 10%), and qualifications: A levels/AS levels or equivalent were all negatively associated with LOAD. These findings suggest that the above traits may have critical impacts on LOAD.

Moreover, some novel potential therapeutic targets of AD were identified as follows: CD33 on Monocytic Myeloid-Derived Suppressor Cells, CD33 on CD33+ HLA DR+ CD14dim, CD33 on CD33+ HLA DR+, CD33 on CD33+ HLA DR+ CD14-, CD33 on CD33dim HLA DR -, CD33 on CD33dim HLA DR+ CD11b-, and Myeloid cell surface antigen CD33 were positively associated with all the three main outcome traits of AD and the risk of LOAD. It has been reported that CD33 is a 67 kDa glycosylated transmembrane protein, a member of the sialic acid-binding immunoglobulin like lectins family (SIGLECS family), which is an important receptor for cell growth and survival, as well as a critical receptor for the clathrin-independent endocytosis pathway and the innate and adaptive immune system functions. CD33 is mainly expressed in microglia, which are a type of glial cells in the central nervous system^[67]. Meanwhile, the splicing efficiency of CD33 affects microglia activation^[68]. Several genome-wide association studies have demonstrated that CD33 is a high-risk gene for AD^[69-70]. In animal models, knockdown of CD33 significantly reduced amyloid plaque levels and knockout mice did not exhibit other health defects. Sialylated glycoproteins and glycolipids on amyloid plaques bind to CD33, which is most likely the cause of the amyloid "immune escape"^[71].

Furthermore, polymorphisms in CD33 can increase the risk of AD by causing neuronal degeneration in the hippocampal and parahippocampal regions of the brain^[72]. Downregulation of the sialic acid-binding domain of CD33 can reduce the risk of developing AD. Therefore, inhibiting CD33 is an effective approach to inhibit the development of AD, and the sialic acid-binding site on CD33 is a promising pharmacophore^[73].

Tubulin-specific chaperone A (TBCA) was negatively associated with all the three main outcome traits of AD, as well as EOAD and LOAD. TBCA is an important member of the tubulin-specific chaperones (TBCs) family. Tian et al. and Nolasco et al. demonstrated that TBCA can regulate the proportion of α and β -tubulin, enabling them to correctly aggregate into cellular microtubules^[74]. Cellular microtubules play important roles in many biological functions, especially in cell movement, cell division, intracellular transport, and cell structure. After silencing TBCA, abnormal microtubule aggregation occurs in mammalian cells, and the cells cannot grow and divide normally, ultimately leading to apoptosis^[75,76]. Moreover, studies have shown that TBCA plays a crucial role in correct β -tubulin folding and α/β -tubulin heterodimer formation^[77]. Protein misfolding can lead to many diseases, such as neurodegenerative diseases. Additionally, higher levels of TBCA are significantly associated with lower AD risk^[78]. These findings suggest that TBCA may serve as a potential protective factor against AD.

Vacuolar protein sorting-associated protein 29 (VPS29) was negatively associated with all the three main outcome traits of AD, as well as EOAD and LOAD. VPS29 is a component of the retromer complex and is highly expressed in the brain, heart, and kidneys, playing an essential role in retromer functions such as synaptic transmission, survival, and movement^[79]. Retromer mainly consists of the VPS26-VPS29-VPS35 trimer and Sorting Nexins (SNXs), and its defects are closely related to various human diseases, including neurodegenerative diseases^[79]. Studies have reported that VPS29 knockdown leads to reduced levels of VPS35 and VPS26^[80,81], which regulates the localization of retromer within neurons and is essential for the aging nervous system^[79]. The retromer complex has been found to regulate the transport of a variety of

substances, including amyloid precursor protein (APP), β -secretase, and phagocytic receptors on microglia. The retromer complex regulates the production of amyloid- β ($A\beta$) by regulating the transport of relevant carrier proteins, thus playing a role in AD^[82]. When the retromer complex malfunctions, the pathway for the reverse transport of APP and β -secretase to the trans-Golgi network is disrupted, resulting in an increase in the production of $A\beta$, which accelerates the pathological process of AD^[83]. Meanwhile, the reduction of phagocytic receptors on the surface of microglia weakens the clearance and protective functions of microglia. Recent studies have shown that stabilizing the retromer complex through chaperone proteins can limit the amyloid processing of APP to reduce the production of $A\beta$ ^[82]. These findings suggest that the retromer complex can serve as a new therapeutic target to intervene in the pathological progression of AD.

Guanine nucleotide-binding protein G(k) subunit alpha (GNAI3) was negatively associated with the three main outcome traits of AD and the risk of LOAD. G proteins are a class of signal transduction proteins that can bind with guanosine diphosphate (GDP) and have guanosine triphosphate (GTP) hydrolysis activity; they have more than 40 types, consisting of alpha, beta, and gamma subunits with a total molecular weight of about 100 kDa, with the alpha subunit having the greatest variation and determining the specificity of the G proteins^[84]. G proteins are intracellular membrane proteins that shuttle between receptors and effector proteins, acting as signal transducers and playing an absolute dominant role in transmembrane cell signaling in the body. All cellular activities are related to signals, and signals are the initiating factors of all cell activities, while physiological responses are only the final results of signals acting on cells. After receiving external stimuli, cells respond by implementing signal transduction through a set of specific mechanisms to ultimately regulate the expression of specific genes, and the whole process is referred to as a cellular signaling pathway. In the pathogenesis of AD, the abnormal content and distribution of multiple signaling molecules, as well as the abnormality of signal transmission pathways, play an important role in AD pathological changes^[85], suggesting that gaining insights into signal transduction mechanisms may provide a potential new pathway to explore the pathogenesis of AD.

Proteasome activator complex subunit 1 (PSME1) was negatively associated with all the three main outcome traits of AD and the risk of LOAD. PSME1 is the encoding gene of the 11s proteasome activator subunit (also known as PA28 α) and is located on human chromosome 14q11.2. PA28 α is an activator of proteasome, which mainly increases the protein degradation activity of 20S proteasome and participates in MHC-I (major histocompatibility complex I) restricted antigen presentation^[86]. Studies have shown that PA28 α overexpression in the brain of female mice can effectively prevent protein aggregation in the hippocampus, thereby reducing depression-like behavior and enhancing learning and memory ability^[87]. Related studies have shown that proteasome function and PA28 α expression are inhibited in the brain of diabetic rats^[87]. The PA28 expression in the diabetic brain has a certain regulatory effect on protein metabolism caused by oxidative damage^[87]. As suggested above, PSME1 may be a new potential therapeutic target for AD and deserves further investigation.

Conclusions

To the best of our knowledge, this is the largest and most comprehensive study to provide important insight into genetic etiology underlying AD based on hypothesis-free Mendelian randomization analysis. In the meantime, we developed the first MR platform for AD, of great clinical and scientific significance that provided a thorough and comprehensive evaluation with regard to risk or protective factors for AD. It also provided physicians and scientists with a very convenient, free as well as user-friendly tool for further scientific investigation. It is important to notice that we recognized CD33, TBCA, VPS29, GNAI3, and PSME1 as novel potential therapeutic

targets for AD that deserve further investigation in more detail. Moreover, this platform currently only covers results from European populations. In the future, we will integrate data from more populations and continuously update new advances in AD research.

Data Availability

All data produced are available online at MRAD platform, which can be freely accessed online at <https://gwasmrad.com/mrad/>. The main project development repository: <https://github.com/ZhaoTianyu-zty/MRAD>.

<https://gwasmrad.com/mrad/>

<https://github.com/ZhaoTianyu-zty/MRAD>

Additional information

Declaration

Ethics approval and consent to participate

Not applicable. All data in this study are sourced from publicly available datasets.

Consent for publication

Not applicable. All data in this study are sourced from publicly available datasets.

Competing interests

We have no conflict of interest to declare. The funders had no role in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript; or in the decision to publish the results.

Funding

This work was supported by the National Natural Science Foundation of China (No. 82302872); the Changchun Science and Technology Planning Project (No. 21ZY18).

Authors' contributions

Zhao TY: Methodology, formal analysis, data curation, visualization, and writing—original draft preparation; Li H: software; Zhang MS, Xu Y and Zhang M: writing—editing; Chen L: conceptualization and supervision. All authors have read and approved the published version of the manuscript.

Author Biographies

Tianyu Zhao is a PhD student in College of Basic Medical Science, Jilin University. Her research interests include bioinformatics and structural biology. Email: zhaoty22@mails.jlu.edu.cn.

Hui Li is a PhD student in Xuanwu Hospital, Capital Medical University. Her research interests include neurology and biology. Email: 3391033864@qq.com.

Meishuang Zhang is a lecturer in School of Nursing, Jilin University. Her research interests include pharmacology and epidemiology. Email: zhangmeishuang@jlu.edu.cn.

Yang Xu is a postgraduate student in College of Basic Medical Science, Jilin University. His research interests include pharmacology and bioinformatics. Email: yxu22@mails.jlu.edu.cn.

Ming Zhang is a professor in College of Basic Medical Science, Jilin University. Her research interests include pharmacology and biology. Email: zhangming_00@126.com.

Li Chen is a professor in College of Basic Medical Science, Jilin University. Her research interests include epidemiology and systems biology. Email: chenl@jlu.edu.cn.

Acknowledgements

We would like to thank Taylor & Francis (<https://www.tandefeditingservices.com/>) for English language editing.

Availability of data and material

Data availability

Publicly available datasets were analyzed in this study. These data can be found here: [MRC IEU OpenGWAS] at (<https://gwas.mrcieu.ac.uk/>), and [UniProt] at (<https://www.uniprot.org/>), the above database search was completed on January 30, 2023.

Code availability

The MRAD platform can be freely accessed online at <https://gwasmrad.com/mrad/>.

The main project development repository: <https://github.com/ZhaoTianyu-zty/MRAD>.

Abbreviations

- AD: Alzheimer's disease
- APP: amyloid precursor protein
- A β : amyloid- β
- CI: confidence interval
- EOAD: early-onset Alzheimer's disease
- EqtL: expression Quantitative Trait Loci
- G protein: guanine nucleotide-binding protein G(k) subunit alpha
- GDP: guanosine diphosphate
- Gln: glutamine
- GTP: guanosine triphosphate
- GWAS: genome-wide association study
- HLA: human leukocyte antigen
- IVW: Inverse Variance Weighted
- LD: linkage disequilibrium
- LDL: low density lipoprotein
- LOAD: late-onset Alzheimer's disease
- MR: Mendelian randomization

- MRAD: Mendelian randomization for Alzheimer's disease
- OR: odds ratio
- PSME1: proteasome activator complex subunit 1
- RCT: randomized controlled trial
- SIGLECS: sialic acid-binding immunoglobulin like lectins
- SNPs: Single nucleotide polymorphisms
- SNXs: Sorting Nexins
- TBCA: tubulin-specific chaperone A
- TBCs: tubulin-specific chaperones
- VLDL: very low density lipoprotein
- VPS29: vacuolar protein sorting-associated protein 29
- WHO: World Health Organization

Highlights

1. To the best of our knowledge, this is the largest and most comprehensive study to provide important insight into genetic etiology underlying AD based on hypothesis-free Mendelian randomization analysis. We generated 400,274 data entries in total, among which the major analysis method of IVW model consists of 73,129 records with 4840 exposure traits, which fall into 10 categories: Disease (n=17,168), Medical laboratory science (n=15,416), Imaging (n=4,896), Anthropometric (n=4,478), Treatment (n=4,546), Molecular trait (n=17,757), Gut microbiota (n=48), Past history (n=668), Family history (n=1,114), and Lifestyle trait (n=7,038).
2. It is also important to note that we developed the first MR platform for AD, of great clinical and scientific significance that provided a thorough and comprehensive evaluation with regard to risk or protective factors for AD. It also provided physicians and scientists with a very convenient, free as well as user-friendly tool for further scientific investigation. The overall method used to construct this platform can be applied to the research of other diseases' etiology.
3. It is also worth noting that we identified CD33, TBCA, VPS29, GNAI3, and PSME1 as novel potential therapeutic targets, which might be promising drug targets for AD and warrant further clinical investigation, especially GNAI3 and PSME1.

References

- [1] J G., J J., D P., et al. (2016) **Therapeutic strategies for Alzheimer's disease in clinical trials** *Pharmacol Rep* **68**
- [2] Ransohoff R.M, El Khoury J (2015) **Microglia in Health and Disease** *Cold Spring Harb. Perspect. Biol* **8**
- [3] Zou Z., Liu C., Che C., et al. (2014) **Clinical genetics of Alzheimer's disease** *BioMed Res. Int*
- [4] Gbd 2019 Dementia Forecasting Collaborators (2022) **Estimation of the global prevalence of dementia in 2019 and forecasted prevalence in 2050: an analysis for the Global Burden of Disease Study 2019** *Lancet Public Health* **7**:e105–e125
- [5] 2022 (2022) **Alzheimer's disease facts and figures** *Alzheimers Dement* **18**:700–789
- [6] World Alzheimer Report 2022 **Life after diagnosis: Navigating treatment, care and support**
- [7] Falsetti L., Viticchi G., Buratti L., et al. (2018) **Interactions between Atrial Fibrillation, Cardiovascular Risk Factors, and Apoe Genotype in Promoting Cognitive Decline in Patients with Alzheimer's Disease: A Prospective Cohort Study** *J. Alzheimers Dis* **62**:713–725
- [8] Femminella G.D., Taylor-Davies G., Scott J., et al. (2018) **Do Cardiometabolic Risk Factors Influence Amyloid, Tau, and Neuronal Function in APOE4 Carriers and Non-Carriers in Alzheimer's Disease Trajectory?** *J. Alzheimers Dis.* **64**:981–993
- [9] Pegueroles J., Jiménez A., Vilaplana E., et al. (2018) **Obesity and Alzheimer's disease, does the obesity paradox really exist? A magnetic resonance imaging study** *Oncotarget* **9**:34691–34698
- [10] Anstey K.J., Cherbuin N., Herath P.M (2013) **Development of a new method for assessing global risk of Alzheimer's disease for use in population health approaches to prevention** *Prev. Sci* **14**:411–421
- [11] Jayaraman A., Pike C.J (2014) **Alzheimer's disease and type 2 diabetes: multiple mechanisms contribute to interactions** *Curr. Diab. Rep* **14**
- [12] Vagelatos N.T., Eslick G.D (2013) **Type 2 diabetes as a risk factor for Alzheimer's disease: the confounders, interactions, and neuropathology associated with this relationship[J]** *Epidemiol. Rev* **35**:152–160
- [13] Cooper J., Pastorello Y., Slevin M (2023) **A meta-analysis investigating the relationship between inflammation in autoimmune disease, elevated Crp, and the risk of dementia** *Front. Immunol* **14**
- [14] Xu Q., Zhang Y., Zhang X., et al. (2020) **Medium-chain triglycerides improved cognition and lipid metabolomics in mild to moderate Alzheimer's disease patients with APOE4-/-: A double-blind, randomized, placebo-controlled crossover trial** *Clin. Nutr* **39**:2092–2105

- [15] Zhu Y., Liu X., Zhu R., et al. (2022) **Lipid levels and the risk of dementia: A dose-response meta-analysis of prospective cohort studies** *Ann. Clin. Trans. Neurol* **9**:296–311
16. [16] Lopes Da Silva, S., Vellas, B., Elemans, S., et al. (2014) **Plasma nutrient status of patients with Alzheimer's disease: Systematic review and meta-analysis** *Alzheimers Dement* **10**:485–502
- [17] Yu J.T., Xu W., Tan C.C., et al. (2020) **Evidence-based prevention of Alzheimer's disease: systematic review and meta-analysis of 243 observational prospective studies and 153 randomised controlled trials** *J. Neurol. Neurosurg. Psychiatry* **91**:1201–1209
18. [18] Douaud, G., Refsum, H., De Jager, C.A., et al. (2013) **Preventing Alzheimer's disease-related gray matter atrophy by B-vitamin treatment** *Proc. Natl. Acad. Sci.* **110**:9523–9528
- [19] Kivipelto M., Mangialasche F., Ngandu T (2018) **Lifestyle interventions to prevent cognitive impairment, dementia and Alzheimer disease** *Nat. Rev. Neurol* **14**:653–666
- [20] Xu W., Tan L., Wang H.F., et al. (2015) **Meta-analysis of modifiable risk factors for Alzheimer's disease** *J. Neurol. Neurosurg. Psychiatry* **86**:1299–1306
- [21] Livingston G., Sommerlad A., Orgeta V., et al. (2017) **Dementia prevention, intervention, and care** *Lancet* **390**:2673–2734
- [22] Livingston G., Huntley J., Sommerlad A., et al. (2020) **Dementia prevention, intervention, and care: 2020 report of the Lancet Commission** *Lancet* **396**:413–446
- [23] Hu Y., Zhang Y., Zhang H., et al. (2022) **Mendelian randomization highlights causal association between genetically increased C-reactive protein levels and reduced Alzheimer's disease risk** *Alzheimers Dement* **18**:2003–2006
- [24] Williams D.M., Finan C., Schmidt A.F., et al. (2020) **Lipid lowering and Alzheimer disease risk: A mendelian randomization study** *Ann. Neurol* **87**:30–39
- [25] Cui G., Li S., Ye H., Yang Y., Huang Q., Chu Y., Shi Z., Zhang X (2022) **Are neurodegenerative diseases associated with an increased risk of inflammatory bowel disease? A two-sample Mendelian randomization study** *Frontiers in immunology* **13**
- [26] Hingorani A., Humphries S (2005) **Nature's randomised trials** *Lancet* **366**:1906–1908
- [27] Emdin C. A., Khera A. V., Kathiresan S (2017) **Mendelian Randomization** *Jama* **318**:1925–1926
- [28] Smith Davey, and Hemani G. (2014) **Mendelian randomization: genetic anchors for causal inference in epidemiological studies** *Hum. Mol. Genet* **23**:R89–98
- [29] Davies N. M., Holmes M. V., Davey Smith G (2018) **Reading Mendelian randomisation studies: a guide, glossary, and checklist for clinicians** *Bmj (Clinical research ed)* **362**
- [30] Elsworth B., Lyon M., Alexander T., et al. (2020) **The Mrc Ieu Opengwas data infrastructure** *bioRxiv*
- [31] Consortium T Uniprot (2018) **UniProt: the universal protein knowledgebase** *Nucleic Acids Res* **46**

- [32] Chen T., Zhang H., Liu Y., et al. (2021) **EVenn: Easy to create repeatable and editable Venn diagrams and Venn networks online** *J. Genet. Genomics* **48**:863–866
- [33] R: The R Project for Statistical Computing [Eb/Ol] **R: The R Project for Statistical Computing [Eb/Ol]**. [2022-02-04]. <https://www.r-project.org/>.
- [34] Bowden J., Del Greco M.F., Minelli C., et al. (2017) **A framework for the investigation of pleiotropy in two-sample summary data Mendelian randomization** *Stat. Med* **36**:1783–1802
- [35] Burgess S., Thompson S.G. (2017) **Interpreting findings from Mendelian randomization using the Mr-Egger method** *Eur. J. Epidemiol* **32**:377–389
- [36] Slob E.A.W., Groenen P.J.F., Thurik A.R., et al. (2017) **A note on the use of Egger regression in Mendelian randomization studies** *Int. J. Epidemiol* **46**:2094–2097
- [37] Bowden J., Davey Smith G., Haycock P.C., et al. (2016) **Consistent Estimation in Mendelian Randomization with Some Invalid Instruments Using a Weighted Median Estimator** *Genet. Epidemiol* **40**:304–314
- [38] Xue H., Shen X., Pan W. (2021) **Constrained maximum likelihood-based Mendelian randomization robust to both correlated and uncorrelated pleiotropic effects** *Am. J. Human Genet* **108**:1251–1269
- [39] Jp H, Sg T, Jj D, et al. (2003) **Measuring inconsistency in meta-analyses** *Bmj* **327**
- [40] Bowden J., Davey Smith G., Burgess S. (2015) **Mendelian randomization with invalid instruments: effect estimation and bias detection through Egger regression** *Int. J. Epidemiol* **44**:512–525
- [41] Hemani G., Zheng J., Elsworth B., et al. (2018) **The Mr-Base platform supports systematic causal inference across the human phenome** *eLife* **7**
- [42] Shiny [Eb/Ol]. [2023-03-11] **Shiny [Eb/Ol]**. [2023-03-11]. <https://shiny.rstudio.com/>.
- [43] Picard C., Nilsson N., Labonté A., et al. (2022) **Apolipoprotein B is a novel marker for early tau pathology in Alzheimer's disease** *Alzheimers Dement* **18**:875–887
- [44] Caramelli P., Nitrini R., Maranhão R., et al. (1999) **Increased apolipoprotein B serum concentration in Alzheimer's disease** *Acta Neurol. Scand* **100**:61–63
- [45] Ym K, Mr E, Cl B, et al. (1998) **Elevated low-density lipoprotein in Alzheimer's disease correlates with brain abeta 1-42 levels** *Biochem. Biophys. Res. Commun* **252**
- [46] Av R, Z R, H K, et al. (2006) **Association between apolipoprotein E polymorphism and serum lipid and apolipoprotein levels with Alzheimer's disease** *Neurosci. Lett* **408**
- [47] R T, S G, P-G M, et al. (2009) **Three-dimensional colocalization analysis of plasma-derived apolipoprotein B with amyloid plaques in App/PS1 transgenic mice** *Histochem. Cell. Biol* **131**
- [48] Takechi R., Galloway S., Pallegage-Gamarallage M.M.S., et al. (2010) **Differential effects of dietary fatty acids on the cerebral distribution of plasma-derived apo B lipoproteins with amyloid-beta** *Brit J. Nutr* **103**:652–662

- [49] Liu Y., Zhong X., Shen J., et al. (2020) **Elevated serum Tc and Ldl-C levels in Alzheimer's disease and mild cognitive impairment: A meta-analysis study** *Brain Res* **1727**
- [50] Wu Y., Wang Z., Jia X., et al. (2019) **Prediction of Alzheimer's disease with serum lipid levels in Asian individuals: a meta-analysis** *Biomarkers* **24**:341–351
- [51] Wang P., Zhang H., Wang Y., et al. (2020) **Plasma cholesterol in Alzheimer's disease and frontotemporal dementia** *Transl. Neurosci* **11**:116–123
- [52] Zhou Z., Liang Y., Zhang X., et al. (2020) **Low-Density Lipoprotein Cholesterol and Alzheimer's Disease: A Systematic Review and Meta-Analysis** *Front. Aging Neurosci* **12**
- [53] Nordestgaard L.T., Christoffersen M., Frikke-Schmidt R (2022) **Shared Risk Factors between Dementia and Atherosclerotic Cardiovascular Disease** *Int. J. Mol. Sci* **23**
- [54] Wingo A.P., Vattathil S.M., Liu J., et al. (2022) **Ldl cholesterol is associated with higher Ad neuropathology burden independent of Apoe** *J. Neurol. Neurosurg. Psychiatry* **93**:930–938
- [55] Hartmann T (2001) **Cholesterol, A beta and Alzheimer's disease** *Trends Neurosci* **24**:S45–48
- 56. [56] Tynkkynen, J., Chouraki, V., Van Der Lee, S.J., et al. (2018) **Association of branched-chain amino acids and other circulating metabolites with risk of incident dementia and Alzheimer's disease: A prospective study in eight cohorts** *Alzheimer Dement* **14**:723–733
- [57] Hashemi M., Banerjee S., Lyubchenko Y.L (2022) **Free Cholesterol Accelerates Aβ Self-Assembly on Membranes at Physiological Concentration** *Int. J. Mol. Sci* **23**
- [58] Treiber-Held S., Distl R., Meske V., et al. (2003) **Spatial and temporal distribution of intracellular free cholesterol in brains of a Niemann-Pick type C mouse model showing hyperphosphorylated tau protein. Implications for Alzheimer's disease** *J. Pathol* **200**:95–103
- [59] Zarrouk A., Debbabi M., Bezine M., et al. (2018) **Lipid Biomarkers in Alzheimer's Disease** *Curr. Alzheimer Res* **15**:303–312
- [60] Zuin M., Cervellati C., Trentini A., et al. (2021) **Association between Serum Concentrations of Apolipoprotein A-I (Apoa-I) and Alzheimer's Disease: Systematic Review and Meta-Analysis** *Diagnostics* **11**
- [61] Tong J.H., Gong S.Q., Zhang Y.S., et al. (2022) **Association of Circulating Apolipoprotein Ai Levels in Patients with Alzheimer's Disease: A Systematic Review and Meta-Analysis** *Front. Aging Neurosci* **14**
- [62] Button E.B., Boyce G.K., Wilkinson A., et al. (2019) **Apoa-I deficiency increases cortical amyloid deposition, cerebral amyloid angiopathy, cortical and hippocampal astrogliosis, and amyloid-associated astrocyte reactivity in App/PS1 mice** *Alzheimers Res. Ther* **11**
- [63] Cannon-Albright L.A., Foster N.L., Schliep K., et al. (2019) **Relative risk for Alzheimer disease based on complete family history** *Neurology* **92**:e1745–e1753
- [64] Wang Y., Wang Q., Li J., et al. (2019) **Glutamine Improves Oxidative Stress through the Wnt3a/β-Catenin Signaling Pathway in Alzheimer's Disease In Vitro and In Vivo** *BioMed Res. Int* **2019**

- [65] Letenneur L., Commenges D., Dartigues J.F., et al. (1994) **Incidence of dementia and Alzheimer's disease in elderly community residents of south-western France** *Int. J. Epidemiol* **23**:1256–1261
- [66] B G Y S, Tk T, et al. (1994) **Influence of education and occupation on the incidence of Alzheimer's disease** *Jama* **271**
- [67] Illegas-Llerena C., Phillips A., Garcia-Reitboeck P., et al. (2016) **Microglial genes regulating neuroinflammation in the progression of Alzheimer's disease** *Curr. Opin. Neurobiol* **36**:74–81
- [68] Malik M., Simpson J.F., Parikh I., et al. (2013) **CD33 Alzheimer's risk-altering polymorphism, CD33 expression, and exon 2 splicing** *J. Neurosci.* **33**:13320–13325
- [69] Hollingworth P., Harold D., Sims R., et al. (2011) **Common variants at ABCA7, MS4A6a/MS4A4e, EPHA1, CD33 and CD2ap are associated with Alzheimer's disease** *Nat. Genet.* **43**:429–435
- [70] Gu X, Dou M, Cao B, Jiang Z, Chen Y (2022) **Peripheral level of CD33 and Alzheimer's disease: a bidirectional two-sample Mendelian randomization study** *Transl Psychiatry* **12**
- [71] A S-P A G, Ar P, et al. (2013) **Alzheimer's disease risk gene CD33 inhibits microglial uptake of amyloid beta** *Neuron* **78**
- [72] Wang W.Y., Liu Y., Wang H.F., et al. (2017) **Impacts of CD33 Genetic Variations on the Atrophy Rates of Hippocampus and Parahippocampal Gyrus in Normal Aging and Mild Cognitive Impairment** *Mol. Neurobiol* **54**:1111–1118
- [73] Miles L.A., Hermans S.J., Crespi G.A.N., et al. (2019) **Small Molecule Binding to Alzheimer Risk Factor CD33 Promotes Aβ Phagocytosis** *iScience* **19**:110–118
- [74] Nolasco S., Bellido J., Gonçalves J., et al. (2005) **Tubulin cofactor A gene silencing in mammalian cells induces changes in microtubule cytoskeleton, cell cycle arrest and cell death** *Febs Lett* **579**:3515–3524
- [75] Gaitanos T.N., Buey R.M., Díaz J.F., et al. (2004) **Peloruside A does not bind to the taxoid site on beta-tubulin and retains its activity in multidrug-resistant cell lines** *Cancer Res* **64**:5063–5067
- [76] Cormier A., Marchand M., Ravelli R.B.G., et al. (2008) **Structural insight into the inhibition of tubulin by vinca domain peptide ligands** *Embo Rep* **9**:1101–1106
- [77] Bellmunt J., Théodore C., Demkov T., et al. (2009) **Phase Iii trial of vinflunine plus best supportive care compared with best supportive care alone after a platinum-containing regimen in patients with advanced transitional cell carcinoma of the urothelial tract** *J. Clin. Oncol* **27**:4454–4461
- [78] Hillary R.F., Gadd D.A., McCartney D.L., et al. (2022) **Genome-and epigenome-wide studies of plasma protein biomarkers for Alzheimer's disease implicate Tbc1a and TREM2 in disease risk** *Alzheimers Dement* **14**
- [79] Ye H., Ojelade S.A., Li-Kroeger D., et al. (2020) **Retromer subunit, VPS29, regulates synaptic transmission and is required for endolysosomal function in the aging brain** *eLife* **9**

- [80] Fuse A., Furuya N., Kakuta S., et al. (2015) **VPS29-VPS35 intermediate of retromer is stable and may be involved in the retromer complex assembly process** *Febs Lett* **589**:1430–1436
- [81] Jimenez-Orgaz A., Kvainickas A., Nägele H., et al. (2018) **Control of RAB7 activity and localization through the retromer-TBC1D5 complex enables RAB7-dependent mitophagy** *Embo J* **37**:235–254
- [82] Zhang Q.Y., Tan M.S., Yu J.T., et al. (2016) **The Role of Retromer in Alzheimer's Disease** *Mol. Neurobiol* **53**:4201–4209
- [83] Seaman M.N.J (2021) **The Retromer Complex: From Genesis to Revelations** *Trends Biochem. Sci* **46**:608–620
- [84] Heese K (2013) **G proteins, p60trp, and neurodegenerative diseases** *Mol. Neurobiol* **47**:1103–1111
- [85] Fowler C.J., Cowburn R.F., Garlind A., et al. (1995) **Disturbances in signal transduction mechanisms in Alzheimer's disease** *Mol. Cell. Biochem* :287–292
- [86] Adelöf J, Andersson M, Porritt M, et al. (2018) **PA28αβ overexpression enhances learning and memory of female mice without inducing 20s proteasome activity** *Bmc neuroscience* **19**
- 87. Donggui Wu. (2021) **Study on the role of PA28 in protein metabolic pathway of oxidative damage in the brain of diabetic rats** :1–52

Editors

Reviewing Editor

Joon-Yong An

Korea University, Seoul, Korea, the Republic of

Senior Editor

Murim Choi

Seoul National University, Seoul, Korea, the Republic of

Reviewer #1 (Public Review):

Summary:

An online database called MRAD has been developed to identify the risk or protective factors for AD.

Strengths:

This study is a very intriguing study of great clinical and scientific significance that provided a thorough and comprehensive evaluation with regard to risk or protective factors for AD. It also provided physicians and scientists with a very convenient, free as well as user-friendly tool for further scientific investigation.

Weaknesses:

(1) The paper mentions that the MRAD database currently contains data only from European populations, with no mention of data from other populations or ethnicities. Given potential differences in Alzheimer's Disease (AD) across different populations, the limitations of the

data should be emphasized in the discussion, along with plans to expand the database to include data from more racial and geographic regions.

(2) Sufficient information should be provided to clarify the data sources, sample selection, and quality control methods used in the MRAD database. Readers may expect more detailed information about the data to ensure data reliability, representativeness, and research applicability.

(3) While the authors mention that the MRAD database offers interactive visualization interfaces, the paper lacks detailed information on how to interpret and understand these visual results. Guidelines on effectively using these visualization tools to help researchers better comprehend the data are essential.

(4) In the conclusion section of the paper, it is advisable to explicitly emphasize the practical applications and potential clinical significance of the MRAD database. The paper should articulate how MRAD can contribute to the early identification, diagnosis, prevention, and treatment of AD and its potential societal and clinical value more clearly.

(5) Grammar and Spelling Errors: There are several spelling and grammar errors in the paper. Referring to a scientific editing service is recommended.

<https://doi.org/10.7554/eLife.96224.1.sa1>

Reviewer #2 (Public Review):

Summary:

This MR study by Zhao et al. provides a comprehensive hypothesis-free approach to identifying risk and protective factors causal to Alzheimer's Disease (AD).

Strengths:

The study employs a comprehensive, hypothesis-free approach, which is novel over traditional hypothesis-driven studies. Also, causal associations between risk/protective factors and AD were addressed using genetic instruments and analysis.

Major comments:

(1) The authors used the inverse-variance weighted (IVW) model as the primary method and other MR methods (MR-Egger, weighted mean, etc.) for sensitivity analysis. However, each method has its own assumption, and IVW is only robust when pleiotropy and heterogeneity are not severe. Rather than using IVW imprudently across all associations, it would be more appropriate to choose the best MR method for each association based on heterogeneity/Egger intercept tests. This customized approach, based on tests of MR assumption violations, yields more stable and reliable results. For reference, please follow up on work by Milad et al. (EHJ - "Plasma lipids and risk of aortic valve stenosis: a Mendelian randomization study"). This study selected the best MR model for each association based on pleiotropy and heterogeneity tests. Given the large number of tests in this work, I suggest initially screening significant signals using IVW, as done, and then validating the results using multiple MR methods for those signals. It is common for MR estimates from different methods to vary significantly (with some being statistically significant and others not), and in such cases, the MR estimates from the best-fitted model should be trusted and highlighted.

(2) Lines 157-160 mentioned "But to date, AD has been reported as hypothesis-driven MR study based on a single factor, ignoring the potential role of a huge number of other risk factors. Also, due to the high degree of heterogeneity present in AD subtypes, which have different biological and genetic characteristics. Thus, the previous studies cannot offer a

systematic and complete viewpoint.". This statement overlooks a similar study published in Molecular Psychiatry ("A Phenome-wide Association and Mendelian Randomization Study for Alzheimer's Disease: A Prospective Cohort Study of 502,493"), which rigorously assessed the effects of 4171 factors spanning 10 different categories on AD using observational analysis and MR. The authors should revise their statement on the novelty of their study type throughout the manuscript and discuss how their work differs from and potentially strengthens previous studies.

(3) Given the large number of tests, the multiple testing issue is concerning. To mitigate potential false positives, I recommend employing the Bonferroni threshold or FDR. The authors should only interpret exposures that are significant at the Bonferroni threshold.

(4) In the discussion, the authors should interpret or highlight exposures that remain significant after multiple testing corrections.

<https://doi.org/10.7554/eLife.96224.1.sa0>